Deliverable 5: User Authentication & Access Control (Task 5)

1. Objective

To implement Role-Based Access Control (RBAC) and audit logging to ensure that all firmware deployments are initiated, approved, and monitored by authorized users only.

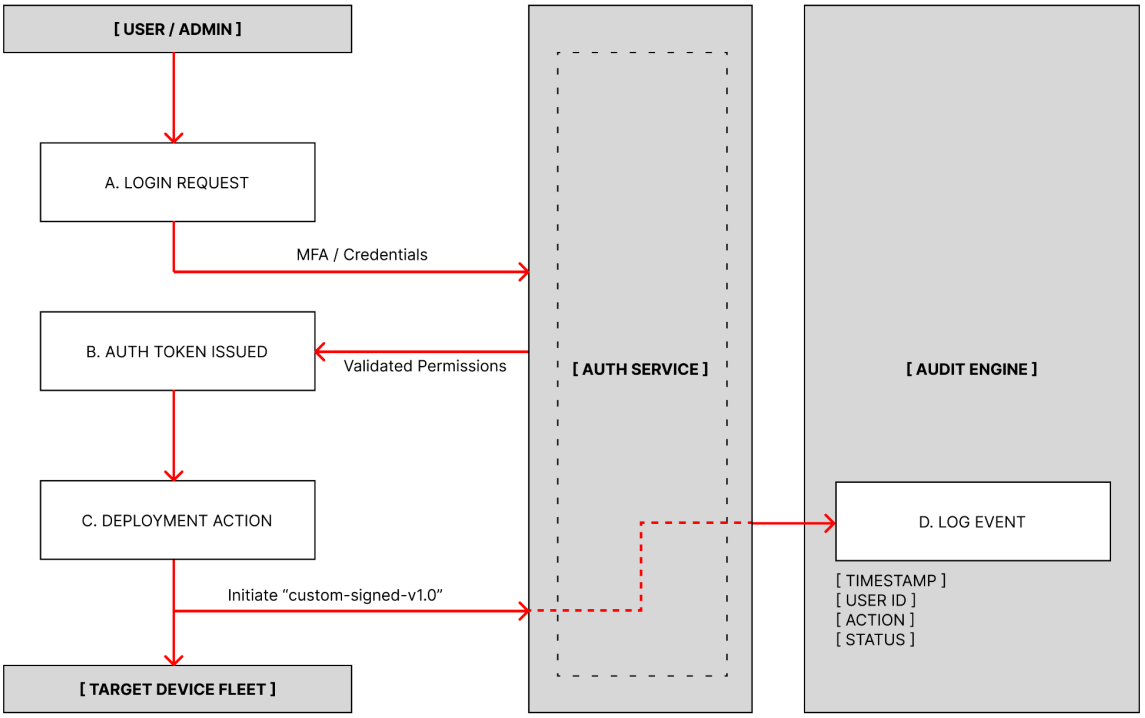
2. RBAC Hierarchy

This blueprint defines the permission layers designed to prevent unauthorized or accidental firmware releases across the IoT fleet.

ROLE-BASED ACCESS CONTROL (RBAC)		
ROLE	PERMISSION LEVEL	RESPONSIBILITIES
Administrator	Full Access	Under Management & Global Configuration
Release Lead	Read/ Write/ Deploy	Signing & Uploading Artifacts
Fleet Operator	Read/ Deploy Only	Scheduling OTA Updates
Auditor	Read-Only	Viewing Deployment History

3. Authentication & Logging Flow

This tracks the security handshake required before the `custom-signed-v1.0` artifact was allowed to reach the production environment.



4. Implementation Details & Constraints

- **Administrative Interface:** Access to the Mender Dashboard is secured via encrypted credentials and session tokens to prevent unauthorized session hijacking.
- **Audit Trail Strategy:** Although the automated "Audit Log" tab is a professional-tier feature, the "Finished" deployments tab serves as a manual audit log, tracking the exact start time, end time, and success/failure status of every artifact deployed.
- **Logging Proof:** Every step of the `custom-signed-v1.0` lifecycle was logged, showing the transition from "Queued" to "Success," providing a clear historical record for compliance.